

Evaluating Multi-Agent Review Workflows for Automated Code Review: A Pre-Registered Study of Decorrelation and Its Limits

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Abstract—Large Language Models are increasingly used for automated code review, and multi-agent frameworks propose dividing the work among specialized agents. It remains unclear whether the *communication topology* among agents—rather than their mere presence, or the extra compute they consume—improves review quality. We compare a four-rung experimental ladder that incrementally introduces additional sampling, role specialization, and peer communication—an Agentless single-LLM baseline, a compute-matched Generalists-3 control that separates “more compute” from “specialization,” a Hierarchical manager-plus-specialists design, and a Decentralized Peer Consensus design—while holding the model, PR snapshot, prompt templates, and evaluation pipeline fixed. Hypotheses and the analysis plan were pre-registered. We report 1,188 reviews over 99 injected-defect PRs, 600 over 50 human-reviewed PRs, and 627 over 30 PRs from a live industrial portal. Role specialization produced no improvement over compute-matched generalist sampling ($p=0.73$), and Agentless achieved the highest F1 among the tested arms (0.49 vs. 0.36–0.38, Holm $p<0.001$); multi-agent workflows raised recall but required 3–9 $\times$ the per-run call budget. Separately, findings produced by three independent model families matched injected ground truth 89% of the time versus 54% for findings recurring across three runs of one model (reproduced under a second pair judge, $\kappa=0.95$). On an unlabeled industrial repository this relationship could not be validated: independent genuineness judges disagreed almost completely ($\kappa=0.03$). Exploratory extensions then trace the limit of the generation axis: the union of *every* configuration we ran—eleven sources spanning seven model families—reaches only ≈ 0.83 recall, and the unreached residue splits into mechanical conventions a deterministic linter recovers and a functional hard-core that an execution probe, not more reading, begins to reach. Multi-agent code review is thus best read as error decorrelation: a precision payoff with a coverage ceiling, where the lever is the right tool per defect class rather than the workflow.

Index Terms—Large Language Models, Multi-Agent Systems, Automated Code Review, Evaluation, LLM-as-a-Judge.

I. INTRODUCTION

Code review gates most software changes, and LLMs now plausibly automate parts of it. A prominent design direction decomposes the task among specialized agents that communicate, as in ChatDev [1] and MetaGPT [2]. Most evaluations, however, compare a *whole* multi-agent system against a single model, conflating two variables: how many agents there are, and how they are organized. A centralized manager may be

cheaper; peer discussion may catch more cross-component issues; or the structure may add cost without accuracy. We ask:

RQ: Holding the underlying model, review target, and evaluation procedure fixed, what incremental effects do additional sampling, role specialization, and inter-agent communication have on the effectiveness and efficiency of automated code review?

Answering this needs a control most studies omit. If a three-specialist team beats a single reviewer, the cause may simply be three samples rather than three *roles*. We therefore add a compute-matched **Generalists-3** arm—the same prompt sampled three times and merged—so specialization is tested against equal compute, not against a single call. Agentless is a strong baseline, since simple non-agent pipelines are competitive on software tasks [3].

Because a benchmark result is only as trustworthy as its evaluation, we evaluate on three complementary datasets— injected-defect ground truth, human-reviewer agreement, and a live industrial repository—and pre-registered the hypotheses, sample sizes, and analysis before collection. Our contributions are:

- A pre-registered experimental ladder separating additional sampling, role specialization, and peer communication through adjacent contrasts, over 2,415 reviews.
- Evidence that the tested homogeneous multi-agent workflows provide no F1 improvement over the single-pass baseline, and that specialization adds nothing over compute-matched sampling.
- Identification of a *cross-model verification* axis where multi-agent value does live—agreement between independent model families predicts golden-set matching far better than repeated sampling of one model.
- An exploratory map of the generation axis’s *limits*: a ≈ 0.83 diff-scoped recall ceiling for an eleven-source union, an anatomy of its residue ($\sim 65\%$ lint-targetable conventions, $\sim 15\%$ a functional hard-core), a deterministic linter that out-recalls the union on its target rules, and an execution probe that recovers functional defects

reading cannot.

- An external-validity boundary: on real unlabeled code that relationship could not be validated, because LLM correctness judges are mutually unreliable ($\kappa=0.03$)—a caution for LLM-judged evaluation generally.

Scope and thread. The registered study isolates the workflow axis, and its answer is largely null; the exploratory extensions (Sections IV-D–IV-F) trace *why*. Every lever we can move reduces to one axis—how *decorrelated* the reviews are—and that axis pays on precision (independent families agreeing on real defects) far more than on coverage, which saturates at a hard ceiling whose residue calls for different tools rather than different workflows. We mark confirmatory and exploratory claims explicitly throughout.

II. RELATED WORK

LLM-based software engineering. SWE-bench established that realistic GitHub tasks are hard for language models [4]. Agentless questions whether autonomous agent systems are necessary at all, showing a simple pipeline is competitive [3]; it is the natural baseline for asking whether multi-agent communication adds measurable value.

Multi-agent software engineering. ChatDev models development as communication among role agents [1]; MetaGPT adds structured workflows to reduce inconsistency [2]. Surveys find the area promising but under-evaluated [5], and recent work catalogs *why* multi-agent LLM systems fail—coordination overhead, cascading errors, weak verification [6]. Concurrent 2026 results point the same way: agent teams can hold individual experts back [11], and mixing different models does not beat the best single model on quality [12]. Our contribution is a controlled experimental ladder that separates the effects of additional sampling, role specialization, and peer communication through adjacent contrasts, and shows where model diversity does pay.

Verification and LLM judges. LLM judges scale evaluation [7], verifier-based selection converts extra samples into quality [8], and diverse judge panels can replace a single judge [9]; but judges show self-preference bias [10] and prompt-form sensitivity [14]. We use judges in two roles—an *objective* same-issue matcher and a *subjective* genuineness judge—and find the first reliable and the second not, which bounds LLM-judged verification and gives a powered test of the correlated-error conjecture for AI code review [13].

III. STUDY DESIGN

A. The four-rung ladder

Each rung adds exactly one capability over its predecessor (Fig. 1): (1) **Agentless**—one LLM reviews the PR in a single pass; (2) **Generalists-3**—the same generalist prompt sampled three times and merged, with no roles and no communication (the compute-matched control that isolates sampling from specialization); (3) **Hierarchical**—a *deterministic* coordinator routes the diff to frontend, backend, and database specialists, whose findings a *deterministic* synthesizer merges;

(4) **Consensus**—the same specialists review independently, then exchange findings and vote.

Coordination and aggregation in the Hierarchical arm are deterministic code, not model calls, so its three LLM calls are exactly its three specialists; only Consensus adds inference stages (nine calls). The model (Claude Haiku 4.5), prompt templates, byte-identical per-PR snapshot, and evaluation pipeline are fixed across arms, but sampling budget, roles, and message flow vary by construction—that is what the ladder manipulates. Inference is therefore not constant across all four arms, so we read *adjacent* contrasts rather than treating topology as a single isolated variable. Each (PR, arm) value is the mean of three runs.

B. Datasets and metrics

E1 — Qodo PR-Review-Bench: 99 PRs with injected defects and explicit ground truth ($4 \times 3 = 1,188$ reviews), supporting objective precision/recall/F1. **E2 — SWE-PRBench:** 50 real PRs with human review comments (600 reviews); metrics measure agreement with reviewers, not absolute correctness. **E3 — industrial:** 30 real merged PRs from an actively developed full-stack portal, with no authoritative ground truth. Its 627 reviews comprise the four-arm ladder ($4 \times 30 \times 3 = 360$) plus three single-pass cross-family arms at three runs each: Haiku 4.5 (90), Kimi K2.5 (90), and GLM 5 (87; GLM returned usable output on 29 of 30 PRs).

Findings are matched to ground truth both strictly (file+line) and *semantically*, where an independent LLM judge makes a binary same-issue decision that rescues near-miss localizations; the judge is threshold-insensitive across 0.5–0.9. Individual malformed findings are dropped and logged rather than failing a whole review, removing a survivorship bias that previously penalized the single-call arm. Operational metrics (latency, tokens, LLM calls, messages) are collected identically everywhere.

For E3, which has no oracle, correctness is replaced by corroboration proxies mirroring the E1 cross-family analysis. *Corroboration depth* clusters independent findings with an LLM pair judge (Nova Pro, $\tau=0.7$, union-find); a cluster’s depth is the number of independent model *families* that produced it. *Proxy precision* is the per-arm rate at which a separate judge rates findings genuine; *proxy recall* is leave-one-out cross-family pool coverage. The industrial axis is cross-*family* (independent models), not cross-*architecture* (one model, different topologies), because only the former carries signal (Section IV-B).

C. Hypotheses and analysis

The design, hypotheses, sample sizes, and analysis plan were pre-registered on OSF and prompts frozen before confirmatory collection. The registered contrasts are summarized with their outcomes in Table I. Two secondary hypotheses were added by a pre-data amendment: **H-verify** (a self-consistency filter lifts each multi-agent arm’s F1 to within 0.02 of Agentless) and **H-hetero-precision** (findings corroborated by ≥ 2 of three independent families {Haiku 4.5, Kimi K2.5,

Figure 1. Four-Rung Experimental Ladder for Comparative Evaluation of Multi-Agent Code Review Topologies

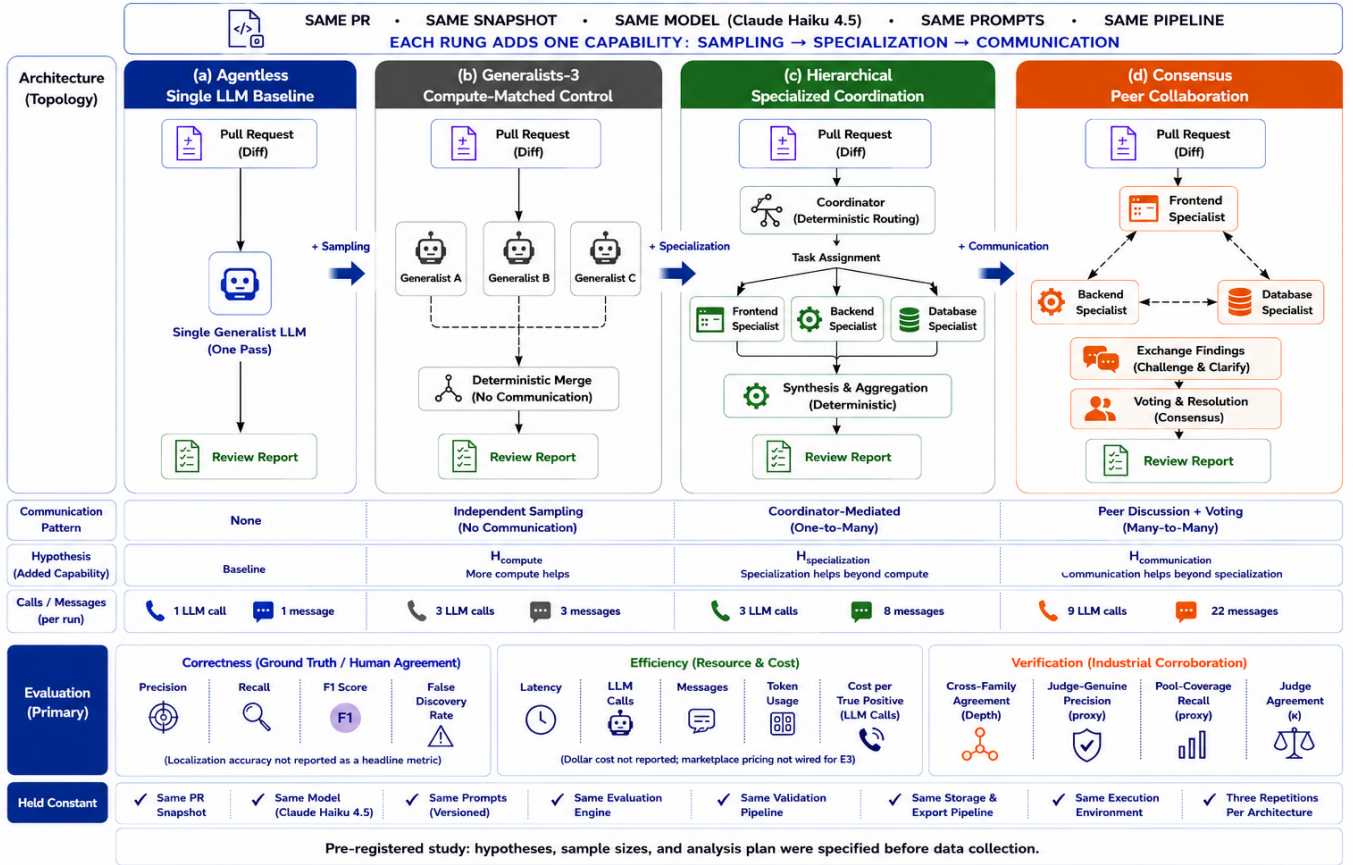


Fig. 1. The four-rung ladder. Each rung adds exactly one capability over its predecessor—sampling, then specialization, then communication—while the PR snapshot, model, prompt templates, and evaluation pipeline are held constant; inference budget necessarily varies with the rung, so we read *adjacent* contrasts rather than treating topology as an isolated variable. Agentless reviews in a single pass; Generalists-3 samples the same generalist three times and merges deterministically without communication (the compute-matched control); Hierarchical routes through a *deterministic* coordinator (one-to-many) and a *deterministic* synthesizer, so its three LLM calls are exactly its three specialists; Consensus adds peer discussion and voting (many-to-many, nine calls). The per-run call and message counts beneath each rung are the coordination overhead quantified in Table II.

GLM 5} gain ≥ 10 precision points at equal-or-higher F1, and all-three agreement exceeds the frozen model’s own three-run recurrence by ≥ 10 points). A further pre-data amendment registered a capability contrast (Section IV-E).

The PR is the unit of analysis. Contrasts use paired Wilcoxon signed-rank tests on per-PR differences, with seeded percentile-bootstrap CIs over those differences and Holm-Bonferroni correction *within* each hypothesis family (families are not pooled). Cliff’s δ is an unpaired dominance measure over the two arms’ per-PR distributions, so it complements rather than matches the paired test: we read significance from Wilcoxon and magnitude from δ . We retain δ rather than a paired rank-biserial correlation because it answers the deployment-relevant question directly—across PRs, how often does one arm dominate the other—and because it is robust to the per-PR averaging over three runs. *Deviation*: the registration specified “Cliff’s δ (paired)”; the implementation computes the standard unpaired dominance form. Since the inferential claims rest on the paired Wilcoxon tests and their bootstrap CIs, and δ is reported only as a magnitude, this

does not affect any registered conclusion, but we report the discrepancy rather than reconcile it silently. Every raw output and judge score is persisted, so all statistics replay deterministically with zero new LLM calls.

IV. RESULTS

A. Generation axis: more agents buy recall, never quality

Table II gives per-arm means over the 99 Qodo PRs together with the cost of obtaining them. The **primary hypothesis is null**: at matched compute, Hierarchical ties Generalists-3 on recall (0.624 vs. 0.626; $p=0.73$, $\delta=-0.005$) at comparable precision. Role specialization contributes nothing beyond sampling—consistent with concurrent evidence that teams average rather than exploit expertise [11]. Nor does compute convert into quality: Agentless dominates every multi-agent arm on semantic F1 (0.487 vs. 0.357–0.378; all Holm $p<0.001$, δ 0.30–0.36). What extra compute buys is recall—Generalists-3 and Hierarchical reach 0.62–0.63 vs. 0.503 (Holm $p<0.001$)—though Consensus, the most expensive arm, does not (0.536, Holm $p=0.12$), leaving H-communication

TABLE I
REGISTERED HYPOTHESES AND CONFIRMATORY OUTCOMES.

Hypothesis	Prediction	Outcome
H-specialization (primary)	Specialists beat compute-matched generalists	Null ($p=0.73$)
H-compute	3 samples beat 1	Supported on recall (0.63 vs. 0.50), not F1
H-communication	Consensus beats Hierarchical	Unsupported (Holm $p=0.12$)
H-efficiency	Cost rises along the ladder	Supported (monotonic)
H-complexity	Effect grows with PR complexity	Registered test unavailable; not supported under an exploratory proxy
H-verify	Self-consistency restores parity	Fails
H-hetero-prec.	Cross-family agreement lifts precision	Confirmed (both sub-claims)

TABLE II
E1 (QODO, 99 PRs): REVIEW QUALITY AND INFERENCE OVERHEAD.
AGENTLESS IS PRECISION- AND F1-DOMINANT; EXTRA AGENTS BUY RECALL AT $3-9\times$ THE CALLS. FDR = $1-\text{PRECISION}$; COST/TP = LLM CALLS PER CONFIRMED TRUE POSITIVE.

Arm	P	R	F1	FDR	Calls	Msgs	Cost/TP
Agentless	0.525	0.503	0.487	0.475	1	1	0.34
Generalists-3	0.270	0.626	0.357	0.730	3	3	0.45
Hierarchical	0.294	0.624	0.378	0.706	3	8	0.50
Consensus	0.322	0.536	0.369	0.678	9	22	1.76

unsupported. Because Consensus and Hierarchical differ in both communication pattern and inference budget (nine calls vs. three), this contrast evaluates the complete peer-consensus *workflow* rather than communication in isolation: the tested workflow did not outperform the hierarchical one despite its larger call and message budget. The trade is stark (Fig. 3a): one confirmed finding costs 0.34 LLM calls under Agentless and 1.76 under Consensus, a $5\times$ premium for a lower-F1 review. The operational ladder rises monotonically in messages, latency, and cost-per-TP, confirming H-efficiency.

On SWE-PRBench the shape replicates in natural data: Generalists-3 raises coverage of human comments (0.68 vs. 0.51, Holm $p=0.001$; Hierarchical 0.63, $p=0.05$), but F1 is a four-way tie (0.325 vs. 0.329–0.357, all n.s., $|\delta|\leq 0.06$). An exploratory Sonnet 4.5 pilot preserves the headline claims—the single-pass F1 lead and the specialization null both hold—though the Consensus arm’s rank *among* the multi-agent arms shifted; the confirmatory robustness arm is deferred (Section V).

Finally, the pre-registered rescue fails. A self-consistency filter (keep findings recurring in ≥ 2 of 3 runs) leaves Hierarchical at 0.378 ($\hat{\Delta}=-0.127$, CI $[-0.179, -0.083]$) and Consensus at 0.373 ($\hat{\Delta}=-0.114$), both entirely below the registered parity band; $k=3$ fails for every arm. A model that agrees with itself is mostly repeating its confident errors—which motivates the next section.

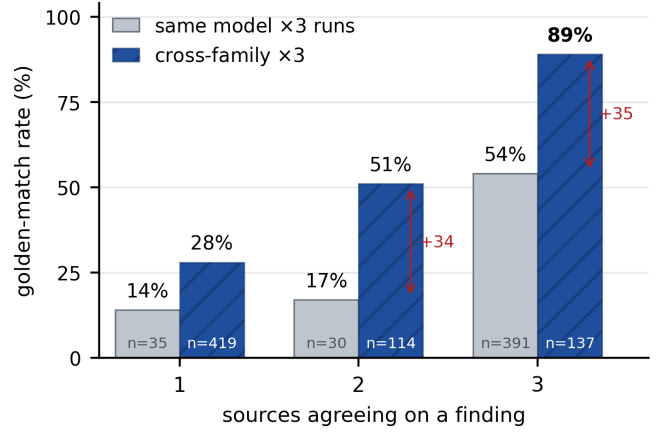


Fig. 2. Agreement predicts truth only when the agreeing sources are *independent*. Repeating one model three times barely improves on a single run until full recurrence, and saturates at that model’s error ceiling (54%); three independent families reach 89%. The gap at equal depth—+34 and +35 points—is the correlated-error mechanism made visible. This gradient does not replicate on unlabeled industrial code (Section IV-H).

B. Verification axis: cross-family agreement is a precision instrument

The registered companion generated agentless reviews with two independent frontier families (Kimi K2.5, GLM 5, both clearing a pre-specified parity gate) over the same PRs with the unchanged frozen prompt; cross-model finding $\leftrightarrow$ finding matching uses a fourth family (Nova Pro) as pair judge, keeping the chain non-circular. The hypothesis was formed on a 20-PR pilot subset, so it is tested on the registered 80-PR disjoint remainder of the 100-instance benchmark (the confirmatory ladder itself uses the 99 instances that survived snapshot validation). There, **both sub-claims hold**. Findings corroborated by ≥ 2 families reach per-PR precision 0.716 vs. 0.575 single-arm ($\hat{\Delta}=+0.141$, CI $[0.094, 0.189]$, $p<10^{-4}$, $\delta=0.364$) at equal-or-higher F1. And findings all three families produce are golden-matched **89%** of the time versus **54%** for findings the frozen model repeats across all three of its own runs (gap +0.348, CI $[0.286, 0.412]$).

Fig. 2 makes the mechanism visible. Cross-family agreement climbs steeply with depth; same-model recurrence stays flat before saturating at its error ceiling. A second run of the same model contributes almost no independent evidence; a second independent family does. The instrument survives a change of pair judge—re-judging all 8,061 candidate pairs with a second pair judge agrees at $\kappa=0.952$ and reproduces the rows.

C. Where the signal lives: functional bugs, not conventions

Cross-family agreement is not uniformly informative—it concentrates on one class of defect. Qodo injects two kinds, and stratifying the 80-PR set by them separates them cleanly (Fig. 3b): universal *functional* bugs are recovered far more often at every depth than project-specific *rule* violations. The asymmetry has a simple cause. A null dereference or an

unguarded input is recognizable from the diff alone, whereas a repository-local convention is invisible to a generic reviewer that has never seen the project’s standards. Independent families converge precisely where the defect is universal.

This shapes what the trust signal is *for*. The ≥ 2 -family set is 52% functional true positives, 20% rule true positives, and 28% golden-unmatched, so cross-family agreement belongs as a high-precision filter for correctness defects—not as a substitute for project-specific linting, which needs repository context rather than more reviewers.

D. The recall axis: a decorrelation ladder with a ceiling (exploratory)

The confirmed cross-family result is a *precision* instrument; on the orthogonal *recall* axis the diversity gain is real but small once compute is matched (Table III). Unioning K temperature-diverse samples of one model (Haiku, $T=0.7$) already reaches 0.61 recall at $K=3$; using distinct model families instead lifts it only to 0.64—a +3-point bonus that saturates by three-to-four families (~ 0.66 at $K=7$). A same-model union at $T=0$ adds essentially nothing (+0.9 points), so comparisons against a $T=0$ baseline overstate the diversity bonus. Pushing decorrelation to its limit, the union of *every* configuration we ran—eleven sources: seven model families, a conditioned “find-new” resampler, and a temperature sweep—reaches only 0.83 recall on the 97-PR set, and leave-one-out shows no single source contributes more than ~ 2 points, so the ceiling is robust. (Nova Pro appears both as a union family and as the pair judge that clusters findings; this mild circularity is confined to this exploratory union—the confirmatory contrast of Section IV-B keeps the judge outside its generators.)

The residue is not random: functional bugs are 89% robustly reachable (covered by ≥ 2 sources), whereas 36% of project-specific rule violations are reached by *no* configuration—reasoning errors decorrelate across sources, but the mechanical-convention blind spot is *correlated*, so eleven draws miss the same violations. Of the unreachable core, $\sim 65\%$ are mechanical conventions a deterministic linter targets and $\sim 15\%$ a functional hard-core (cross-file renames, dynamic imports, races) invisible to a diff-scoped reviewer. Wiring in a linter confirms the division of labour: on the two convention-authored repositories a single checker out-recalls the entire eleven-source union on exactly the rules it targets (0.54 vs. 0.46), and their hybrid reaches 0.61. No precision back-end changes this frontier—neither cross-family agreement (Section IV-B) nor decomposed aspect-based verification lifts F1 above a plain independent union; filters and verifiers trade recall for precision.

E. Neither capability nor generation-side structure closes the gap

Three further levers fail to target the residue. **Capability (registered)**. A pre-data amendment tested whether a stronger model’s advantage concentrates on reasoning-limited functional bugs. On the 80-PR remainder, with single-pass diff-only reviews, the difference-in-differences is null (DiD= -1.8

TABLE III
UNION RECALL BY NUMBER OF DRAWS K (SEMANTIC MATCHING, 97 QODO PRs). HOMO = ONE MODEL (HAIKU) AT $T=0.7$; HETERO = HAIKU PLUS ONE DISTINCT FAMILY PER RUNG (KIMI, GLM, DEEPSEEK, NOVA, LLAMA 4, PALMYRA).

K	1	2	3	4	5	6	7
Homo ($T=0.7$)	0.49	0.58	0.61	—	—	—	—
Hetero (families)	0.51	0.61	0.64	0.65	0.65	0.66	0.66

Diversity bonus (hetero – homo) = +0.03 at $K=2, 3$. Family order affects per-step marginals, not the $K=\max$ total.

points, CI $[-8.6, 5.5]$, $p=0.68$): Sonnet 4.5 lifts both defect types modestly over Haiku (functional 0.70 $\rightarrow$ 0.74, rule 0.32 $\rightarrow$ 0.37)—a broad, undirected lift, not a targeted capability. **Module decomposition (exploratory)**. Reviewing each module with its own agent and unioning beats one whole-PR review by +6.4 recall points ($p=0.016$, $N=50$), but that is the $\sim 2.6\times$ compute, not the structure: at a matched call budget, module-union *trails* plain multi-sampling on the multi-module subset (0.54 vs. 0.57, n.s., $n=24$). **Error-type specialists (exploratory)**. Prompting one model for only conventions and another for only functional bugs suppresses output (5.3 vs. 16.1 findings/PR) and reduces per-type recall (conventions 24% vs. the generalist’s 32%; functional 59% vs. 68%); the two-specialist team reaches recall 0.47, below the generalist (0.50) and the same-budget baselines (multi-sample 0.58, cross-family 0.60). Prompt personas on one model are a *negative* decorrelation source; specialization pays only when it changes the detection *mechanism*—a deterministic checker, an independent family, or execution—not the prompt.

F. Execution begins to reach what reading cannot (exploratory)

The functional hard-core resists reading in every form we tried. Passive context—whole changed files plus one-hop dependencies retrieved at the review commit—does not raise recall over the diff alone (0.37 $\rightarrow$ 0.35, n.s., 50 PRs on an execution-grounded review benchmark [15]), and a read-only repository-navigating agent (158 PRs on the same benchmark) raises precision (0.17 vs. 0.13 diff-only) at *lower* recall—agency verifies and prunes, another precision mechanism, not a coverage lever. A light execution probe shows why: asked to write fully self-contained reproductions (standard library only) for functional findings, the model succeeds on only $\sim 15\%$ ($n=20$) and declines the rest as inseparable from the repository, framework, or runtime—the hard-core is genuinely repo-integrated. Real execution is different. On runtime-error SWE-bench instances whose issue text does not localize the defect, feeding the reviewer the failing-test traceback lifts file-localization recall from 1/6 to 5/6—and from 25% to 100% on the stratum where the traceback names the fix file—whereas on feature-addition instances whose issue already names the component it is a ceiling (both arms 100%). The execution signal recovers exactly the repo-integrated functional defects that diff reading and read-only navigation cannot (small n , single reviewer model; directional) [16]. Together with the

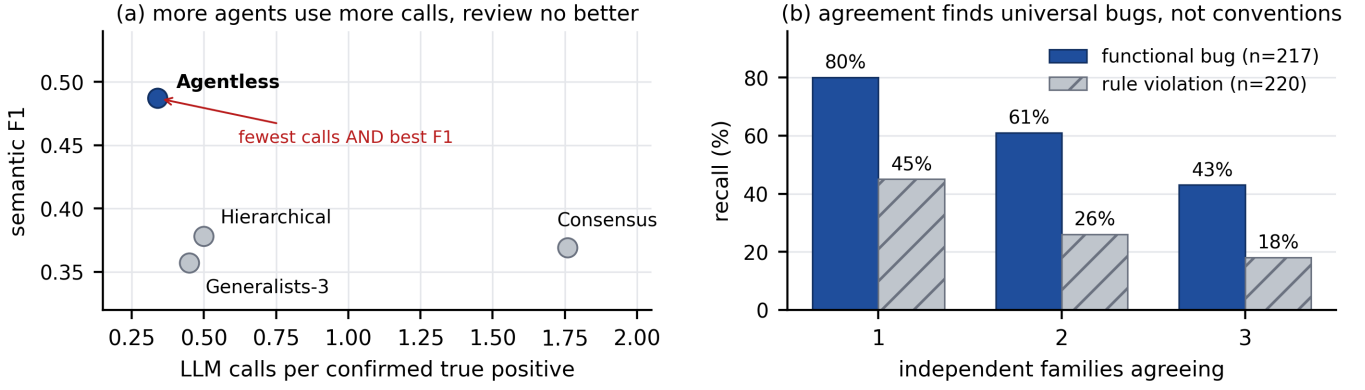


Fig. 3. Two supporting views of the same conclusion. (a) The accuracy/cost frontier: Agentless Pareto-dominates—it is simultaneously the cheapest arm (0.34 LLM calls per confirmed true positive) and the most accurate (semantic F1 0.487), while Consensus costs $5\times$ more for a worse review. (b) Where cross-family agreement carries signal: universal functional bugs are recovered far more often at every depth than project-specific rule violations, which a reviewer without repository context cannot know. Together, the panels show that homogeneous multi-agent workflows increase inference cost without improving F1, while cross-family corroboration is most informative for universal functional defects.

linter result the map is complete: reading buys precision, while a mechanism-changing signal—a deterministic checker, an independent model family, or executed code—is what moves coverage.

G. How complete is the oracle?

Every precision figure above is measured against the injected golden set, so its completeness bounds their interpretation. We audited it confirmatorily: three independent judge families read every golden-*unmatched* finding against the diff (“does this identify a genuine problem?”), plus a calibration pass on golden-*matched* findings. Llama 3.3 calls 76% of unmatched findings real (92% on known defects); Mistral Large 3 calls 73% (90%); DeepSeek V3.2 calls only 33% (71%).

The disagreement is definitional, not a capability gap: DeepSeek is a strict under-caller that rejects roughly 29% of *known injected defects*—largely rule and style violations—while the lenient judges accept warranted concerns. Critically, inter-judge agreement here is poor ($\kappa=0.23$ – 0.58) against the pair judge’s $\kappa=0.95$ on same-issue matching. Two questions that look alike behave differently: whether two findings describe the same issue is objective and stable across judges; whether a finding is a real bug is not settled by an LLM judge of any size.

Three conclusions survive every definition. The golden set is materially incomplete, so golden-referenced precision *understates* all arms equally, leaving relative comparisons intact. The ≥ 2 -family set’s effective precision is bracketed at 86–95%. And false positives are less false than they appear: 21–32% are localization near-misses—real defects at the wrong line—and high-severity findings are almost always real. The magnitude awaits human adjudication; the direction does not. The audit also predicts what follows: remove the golden set, as the industrial arm does, and the only correctness signal left is the one that was already unreliable.

H. Industrial validation: the instrument needs an oracle

We ran the registered industrial arm on 30 real merged PRs (627 reviews: the four-arm ladder plus the cross-family axis), scored with the same instruments but with proxies in place of golden labels. Two patterns transfer. The communication-cost ladder is unchanged (Agentless 1 call / 13s through Consensus 9 calls / 173s), and the pool-coverage proxy gives the same qualitative verdict—more agents do not buy coverage: the single-pass baseline is highest (0.27) and the most-agentful Consensus lowest (0.10). (Pool coverage is a different construct from golden recall—on which Agentless ranked *lowest*—so we compare directions, not ranks.)

The verification instrument, however, could not be *validated* on unlabeled code—the gradient of Fig. 2 cannot be measured here. Cross-family agreement is rare: only 23 findings reach depth-2 and 5 reach depth-3 of 904 clusters, versus 114 and 137 on E1. More decisively, the genuineness judges disagree almost completely—Nova rates 99% of findings genuine (a rubber stamp), DeepSeek 68%, at $\kappa=0.03$ —so the depth→precision relation is set by judge choice, not corroboration: flat at 100% under Nova, and *inverted* under the discriminating judge, where findings families agree on score 0% versus 15% for solo findings. This is precisely the failure Section IV-G anticipated, now with no golden set left to bound it: the objective matcher still works, and the correctness judge does not. The negative result is therefore about the *measurement apparatus*, not the underlying signal—with mutually inconsistent oracles and depth counts of 23 and 5, cross-family agreement can here be neither confirmed nor refuted.

I. Complexity interaction: the effect does not grow with PR complexity

The registered stratifier was PR type, which requires the diff; the confirmatory runs did not persist it, so we substitute the closest available proxy—the number of distinct files an

instance’s defects span—and label this analysis **exploratory**. **No interaction survives on any contrast** (all Holm $p \geq 0.42$, $|\delta| \leq 0.12$, Spearman $|\rho| \leq 0.17$). Complexity degrades every arm by a similar margin (recall $0.52 \rightarrow 0.46$ Agentless, $0.65 \rightarrow 0.56$ Generalists-3 and Hierarchical, $0.57 \rightarrow 0.45$ Consensus) and the ordering is stable, so no topology rescues complex PRs. The one directional hint runs *against* the registered prediction: Consensus minus Hierarchical recall is 0.000 on simple PRs and -0.119 on complex ones (Holm $p=1.0$)—a non-significant counter-directional trend, not a finding.

V. THREATS TO VALIDITY

Construct. E3 has no ground truth, so its metrics are proxies and are labelled as such. Even the benchmark’s golden set is incomplete (Section IV-G), so golden-referenced precision is a lower bound; relative comparisons are unaffected, because incompleteness applies to every arm identically. The pair judge is validated instrument-side ($\kappa=0.95$) but its human validation is pending, so corroboration numbers remain instrument-relative.

Internal. Differences could stem from confounds other than the manipulated capability. We hold the model, prompt templates, evaluation pipeline, and a byte-identical PR snapshot fixed while manipulating sampling budget, role structure, and communication according to the experimental ladder; the compute-matched Generalists-3 control separates specialization from raw sampling, and the evaluation engine replays deterministically from stored artifacts.

External. Qodo is a synthetic-injection benchmark: planted defects are cleaner and more localizable than natural review targets, so absolute scores sit on a favorable setting even though within-benchmark comparisons hold; the SWE-PRBench replication and the industrial arm partly offset this. One registered piece is deferred—the Sonnet 4.5 robustness arm at confirmatory scale, whose cost is prohibitive at $99 \times 4 \times 3$ —so the exploratory pilot stands in its place and single-model-family generalizability, specifically the ranking *within* the multi-agent arms, is stated as a limitation rather than claimed. The exploratory extensions (Sections IV-D–IV-F) are small- n , single-benchmark probes and are labeled as such; the execution probe in particular is directional ($n=6$, one reviewer model). The industrial repository is partly AI-authored, which is disclosed.

Conclusion. The headline claims (the specialization null, Agentless F1 dominance, H-verify’s failure, and both H-hetero sub-claims) are confirmatory: registered sample sizes, paired tests, effect sizes with CIs, and Holm correction within families. The finer stratifications, RQ3, and the ceiling/capability/execution extensions are exploratory and labeled as such; E3’s depth-2/3 counts are small (23 and 5), so those CIs are wide.

VI. CONCLUSION

Under a pre-registered, frozen-model design, none of the tested multi-agent workflows improved automated code review. Role specialization adds nothing over compute-matched

sampling (the primary null), no arm exceeds the single-pass baseline’s F1, extra agents buy recall at a $3\text{--}9\times$ larger per-run call budget, the self-consistency rescue our own pilot suggested does not replicate, and the effect does not grow with PR complexity. The *verification* axis is where multi-agent value lives: agreement between independent model families is a registered precision instrument that survives a change of pair judge—findings all three families produce match injected ground truth 89% of the time versus 54% for one model repeating itself—and the signal concentrates on functional bugs.

That instrument, however, has a precondition. On real unlabeled code we could not validate it, because LLM correctness judges disagree at $\kappa=0.03$ while the objective same-issue matcher stays reliable at $\kappa=0.95$; the industrial result therefore bounds *oracle-based validation*, not cross-family agreement itself, which remains untested outside the labeled setting. The practical guidance follows that boundary: on review tasks resembling the benchmarked ones, do not add homogeneous agents for quality, and treat cross-family agreement as a candidate trust signal to be calibrated against a project’s own oracle before it is relied upon.

Read end to end, the exploratory extensions locate the organizing axis: error decorrelation, which pays in precision and saturates in coverage near 0.83. The residue is a division-of-labour problem, not an architecture problem—lint the mechanical conventions, union-sample independent families for functional bugs, and reserve execution for the hard functional core. Future work is a fixed-definition *human* adjudication—the one oracle these judges could not provide—and scaling the execution probe to confirmatory strength.

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REFERENCES

- [1] C. Qian et al., “ChatDev: Communicative agents for software development,” in *Proc. ACL*, 2024.
- [2] S. Hong et al., “MetaGPT: Meta programming for a multi-agent collaborative framework,” in *Proc. ICLR*, 2024.
- [3] C. S. Xia, Y. Deng, S. Dunn, and L. Zhang, “Agentless: Demystifying LLM-based software engineering agents,” arXiv:2407.01489, 2024.
- [4] C. E. Jimenez et al., “SWE-bench: Can language models resolve real-world GitHub issues?” in *Proc. ICLR*, 2024.
- [5] X. Zhang et al., “LLM-based multi-agent systems for software engineering: Literature review, vision and the road ahead,” arXiv preprint, 2024.
- [6] Y. Li et al., “Why do multi-agent LLM systems fail?” arXiv preprint, 2025.
- [7] L. Zheng et al., “Judging LLM-as-a-judge with MT-Bench and Chatbot Arena,” in *Proc. NeurIPS*, 2023.
- [8] K. Cobbe, V. Kosaraju, M. Bavarian, M. Chen, H. Jun, L. Kaiser, M. Plappert, J. Tworek, J. Hilton, R. Nakano, C. Hesse, and J. Schulman, “Training verifiers to solve math word problems,” arXiv:2110.14168, 2021.
- [9] P. Verga, S. Hofstätter, S. Althammer, Y. Su, A. Piktus, A. Arkhangorodsky, M. Xu, N. White, and P. Lewis, “Replacing judges with juries: Evaluating LLM generations with a panel of diverse models,” arXiv:2404.18796, 2024.

- [10] A. Panickssery, S. R. Bowman, and S. Feng, “LLM evaluators recognize and favor their own generations,” arXiv:2404.13076, 2024.
- [11] A. Pappu, B. El, H. Cao, C. di Nolfo, Y. Sun, M. Cao, and J. Zou, “Multi-agent teams hold experts back,” in *Proc. ICML*, 2026 (arXiv:2602.01011).
- [12] W. Li, Y. Lin, M. Xia, and C. Jin, “Rethinking mixture-of-agents: Is mixing different large language models beneficial?” arXiv:2502.00674, 2025.
- [13] C. Zietsman, “The specification as quality gate: Three hypotheses on AI-assisted code review,” arXiv:2603.25773, 2026.
- [14] M. Sclar, Y. Choi, Y. Tsvetkov, and A. Suhr, “Quantifying language models’ sensitivity to spurious features in prompt design,” in *Proc. ICLR*, 2024 (arXiv:2310.11324).
- [15] Y. Zhang et al., “Code review agent benchmark,” arXiv:2603.23448, 2026.
- [16] R. Wang et al., “SWE-Review: Closing the loop on issue resolution with agentic code review,” arXiv:2607.06065, 2026.